

FULL NAME: _____

BLOCK: _____

Adding & Subtracting Rational Expressions + Resistance DOK-3 (Q#15 Prep)

Lesson 4-4 · Quick · Student Packet

OBJECTIVES

Math Objective	Add and subtract rational expressions with like and unlike denominators; identify the LCM of polynomial denominators; simplify results.
Language Objective	Explain why adding rational expressions requires common denominators, using the word 'LCM' and referring to the analogy with fractions.
Essential Understanding	Rational expressions behave like rational numbers — closed under addition and subtraction, and requiring a common denominator before you can combine them.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency adding/subtracting rational expressions; understanding closure under these operations (Topic 4 assessment Q#15 prep).
Essential Question	When you add two rational expressions with different denominators, why must you find the LCM first?
Materials	Student packet, pencil, calculator. DOK-3 Practice #32 (electrical resistance) is the capstone and rehearses the structural reasoning tested on Topic 4 LEHS Q#15.

DO NOW

BRIDGE SCRIPT (read aloud after Do Now)

"Yesterday (Lesson 4-3) you practiced MULTIPLYING and DIVIDING rational expressions. Today you'll ADD and SUBTRACT them — which requires one new move that multiplication didn't: finding a common denominator first. The Model & Discuss prompt shows two students adding $1/2 + 1/3$ in conflicting ways. Which is right?"

Do Now

CRITIQUE EXPLAIN Teo and Evander find the following exercise in their homework: $(1)/(2) + (1)/(3) + (1)/(9)$

[IMAGE: A tablet displaying the math problem $1/2 + 1/3 + 1/9$]

- A. Teo claims that a common denominator of the sum is $2 + 3 + 9 = 14$. Evander claims that it is $2 \cdot 3 \cdot 9 = 54$. Is either student correct? Explain why or why not.
- B. Find the sum, explaining the method you use.
- C. Construct Arguments Timothy states that the quickest way to find the sum of any two fractions with unlike denominators is to multiply their denominators to find a common denominator, and then rewrite each fraction with that denominator. Do you agree?

LAUNCH

[BOOK] RULES — ADD/SUBTRACT RATIONAL EXPRESSIONS (keep printed on your page)

1. LIKE denominators: combine numerators, keep the denominator, simplify.
2. UNLIKE denominators:
 - (a) factor each denominator,
 - (b) find the LCM,
 - (c) rewrite each fraction with the LCM as denominator,
 - (d) combine numerators,
 - (e) simplify.
3. The LCM of two polynomial denominators includes every DISTINCT factor, taken to its highest power across both.
4. Always check whether the final expression can be simplified by cancelling a common factor.

[WARN] COMPRESSED LESSON — TWO EXAMPLES IN LAUNCH

This is a single-period quick treatment. Teacher models BOTH Example 3 (add unlike denominators) AND Example 4 (subtract unlike denominators) during Launch.

Watch carefully on Example 4 — when you subtract, you must distribute the minus sign across every term of the second numerator. This is the #1 error source.

EXPLORE

Bank-item labels (in order):

1. Try It 3 — Add rational expressions (unlike denominators)
2. Try It 4 — Subtract rational expressions (unlike denominators)
3. Practice #12 — Like denominators
4. Practice #16 — Unlike denominators, factor first
5. Practice #26 — Subtract with sign-distribution trap
6. Practice #32 [STAR] DOK-3 ELECTRICAL RESISTANCE — Topic 4 LEHS Q#15 prep

Try It 3 — Add rational expressions (unlike denominators)

Find the sum.

- a. $((x+6)/(x^2-4) + (2)/(x^2-5x+6))$
- b. $((2x)/(3x+4) + (4x^2-11x-12)/(6x^2+5x-4))$

Try It 4 — Subtract rational expressions (unlike denominators)

Simplify.

- a. $((1)/(3x) + (1)/(6x) - (1)/(x^2))$
- b. $((3x-5)/(x^2-25) - (2)/(x+5))$

Practice #12 — Like denominators

Generalize Explain how addition and subtraction of rational expressions is similar to and different from addition and subtraction of rational numbers.

Practice #16 — Unlike denominators, factor first

Look for Relationships Find the LCM of $3x^2 + 7x + 2$ and $9x + 3$.

Practice #26 — Subtract with sign-distribution trap

On Saturday morning, Ahmed decided to take a bike ride from one end of the 15-mile bike trail to the other end of the bike trail and back. His average speed the first half of the ride was 2 mph faster than his speed the second half. Find an expression for Ahmed's total travel time. If his average speed for the first half of the ride was 12 mph, how long was Ahmed's bike ride?

[IMAGE: Bike trail with cyclist, sign "Trail length 15 miles", one label "Speed $x+2$ " and another label "Speed x ".]

Practice #32 [STAR] DOK-3 ELECTRICAL RESISTANCE — Topic 4 LEHS Q#15 prep

Model With Mathematics The resistance of electrical circuits A and B are shown. Find the ratio of the resistance of Circuit A to Circuit B.

[GRAPH / TIKZ figure]

SHARE / SUMMARY**[SEARCH] BRIDGE TO TOPIC 4 LEHS Q#15**

The resistance move — find LCM, combine reciprocals into a single rational expression — is the structural reasoning tested on Topic 4 LEHS Q#15 (operations/closure). You have now rehearsed it.

Sentence frames

Pair share (Resistance): "For the parallel circuit, $1/R_{\text{total}} = 1/R_1 + 1/R_2$. I found a common denominator of ____, added the reciprocals to get $1/R_{\text{total}} = ____$, then flipped to get $R_{\text{total}} = ____$. The ratio R_A/R_B simplified to ____ because ____."

Whole class: one pair reads their resistance solution. Class signals thumbs up/sideways.

EXIT

To add rational expressions with unlike denominators, I must first find the ____ of the denominators.

Practice #32 (resistance) showed me that reciprocal-sum problems (like $1/R_1 + 1/R_2$) use the same algebraic steps as adding rational expressions because ____.

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

Optional #1 (Savvas Practice #34 — compound-fraction multi-select)

Which of the following compound fractions simplifies to $(x + 1)/(x - 3)$? Select all that apply.

$$((x^2 + 5x + 4)/(x^2 + 2x - 8))/((x^2 - 4x + 3)/(x^2 - 3x + 2))$$

$$((x^2 - 1)/(x^2 - 4))/((x^2 + x - 7)/(x^2 + 5x + 6))$$

$$((x^2 + 3x - 10)/(x^2 - 16))/((x^2 - 4x - 5)/(x^2 - 1))$$

$$((x^2 + 3x - 10)/(x^2 - 5x + 6))/((x^2 - 25)/(x^2 - 4x - 5))$$